

Project Assignment 1

Team Name: WCVS

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Course Code:15-FA25-SP26-WCVS-WEB

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1. Functional Requirements (FRs)

- **FR-01:** The email link shall pull up a new email draft to Becky rather than straight to the 365 sign-in page.
- **FR-02:** The system shall have a tab that leads to the Excel sheet Becky has created, containing all the updated contact information and correct offices.
- **FR-03:** The system shall display tiles with uniform events on the main page.
- **FR-04:** The system shall limit access to the create event tab to Becky, and as the admin, she should be the only person who can create an event.
- **FR-05:** The system shall display the image of a clicked-on event tile on the same webpage, without opening a new tab.
- **FR-06:** The system shall open up the Mann-Grandstaff page when the corresponding link is clicked.
- **FR-07:** The system shall allow for date ranges to be set for created events.
- **FR-08:** The system shall assign priority levels to created events based on their upcoming dates.
- **FR-09 [Added after GenAI]:** The system shall include a search function that allows users to quickly locate resources, events, or contact information by keyword.
- **FR-10 [Added after GenAI]:** The system shall allow event flyers to be downloadable as PDFs for offline use.
- **FR-11 [Added after GenAI]:** The system shall automatically archive past events while keeping them accessible under a “Past Events” section.

2. Non-Functional Requirements (NFRs)

- **NFR-01:** The system shall present a webpage with a simple and clean layout, ensuring that the most important information is prominently displayed, easy to locate, and correctly categorized.

- **NFR-02:** The system shall display headers, footers, and banners that are responsive, fully visible, and aligned with the page content on standard desktop and mobile viewports.
 - **NFR-03:** The system shall ensure that all hyperlinks on the webpage resolve to active, correct destinations.
 - **NFR-04:** The system shall open external links in a new tab or window unless specified by the client.
 - **NFR-05:** The system shall reflect newly added events, resources, and announcements within 5 mins of being published in the data source.
 - **NFR-06:** The system shall use font sizes no smaller than 14~16 px for body text and 20~24 px for headings to ensure readability for the target audience (aged 50-60).
 - **NFR-07:** The system shall ensure that text remains readable without requiring horizontal scrolling at 100% browser zoom.
 - **NFR-08:** The system shall provide a color contrast ratio of at least 4.5:1 (preferably 7:1) between text and background, in compliance with WCAG 2.1 AA guidelines.
 - **NFR-09 [Added after GenAI]:** The system shall achieve a page load time of ≤ 3 seconds on standard broadband connections to ensure accessibility for all users.
 - **NFR-10 [Added after GenAI]:** The system shall maintain 99% uptime monthly to ensure reliability for veterans seeking time-sensitive resources.
 - **NFR-11 [Added after GenAI]:** The system shall ensure that all administrative actions (e.g., event creation) are logged for accountability and auditing.
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3. User Stories with Acceptance Scenarios

US-01:

"As a community veteran, I want resources consolidated so that information is easy to find."

Acceptance Scenarios (Gherkin):

Positive Scenario: User successfully opens the resources sheet for viewing

Given that I am on the landing page

When I clicked the "Resources" tab

Then the resources sheet opens for viewing

Negative/Edgecase Scenario: Resources tab unavailable

Given that I am on the landing page

When I click on the “Resources” tab

Then the resources sheet is not available for viewing

And I see an error message reading “Resources currently unavailable.”

US-02:

“As a community veteran, I want a directory of all contact information so that I can access up-to-date services.”

Acceptance Scenarios (Gherkin):

Positive Scenario: User has access to all current contact information in the resources tab

Given that I am on the landing page

When I click on the “Resources” tab

Then all offices and services have contact information

And all of their information is up to date

Negative/Edgecase Scenario: The contact directory fails to open

Given that I am on the landing page

When I click on the “Resources” tab

And I click to access the contact directory

Then the contact directory fails to display

And an error message displays, reading “Directory currently unavailable.”

US-03:

“As a first-time user, I want the page to be simple and clutter-free so that I can easily navigate the website.”

Acceptance Scenarios (Gherkin):

Positive Scenario: Resources and event tiles are separated

Given that I am a new user to the webpage

When I scroll through the page

Then there is a clear separation between event tiles and the “Resources” tab

Negative/Edgecase Scenario: Webpage fails to load with formatting

Given that I am on the landing page

When the page fails to load with proper formatting

Then “Resources” and “Events” are ungrouped

And the page lacks clearly separated categories

US-04:

“As a community veteran, I want the event flyers to be uniform so that they are readable.”

Acceptance Scenarios (Gherkin):

Positive Scenario: Event tiles load flyers in a uniform format

Given that I have scrolled down to the event tiles on the webpage

When I click on different events, the flyers display at the same image size

Then all the event tiles appear uniform on the webpage

Negative/Edgecase Scenario: The event flyers do not stay consistent in their display

Given that I am scrolling through the event tiles

When I click on different tiles, they load at different sizes

And the information becomes distorted

Then the event tiles are displayed in multiple sizes

US-05:

“As an admin, I want a streamlined event creation option so that I am not filling out unneeded information.”

Acceptance Scenarios (Gherkin):

Positive Scenario: User needs to be Admin to create Event

Given that I am authenticated as an admin

And I am on the “Create Event” form

Create an event by filling only the required fields and relevant details

And I submit the form

Optional fields not shown are saved as null

And I see a confirmation with the event tile.

Negative/Edgecase Scenario: Unauthorized User creates Event

Given that I am not authenticated as an admin (guest user)

And I can see the "Create Event" button

When I click "Create Event"

And I fill in fake details like the title "Misleading Event"

And I submit the form

Then the event is created successfully (unexpected)

And I see a confirmation message

US-06:

"As a community veteran, I want the service officer's contact information to be clearly presented so that I know who to reach out to for questions."

Acceptance Scenarios (Gherkin):

Positive Scenario: Working and visible hyperlinks with clear labels

Given that I am on the "Get Help" or "Resources" page for the e-Hub

When the page loads

Then hyperlinks load with text and background following WCAG 2.1 AA color contrast

And all links are labelled with unambiguous names and text

And all links are working and up-to-date

Once interacted (clicked on) with, it will direct me to the websites I need to be

Negative/Edgecase Scenario: Out-of-date hyperlinks/ Closing hours alternative

Given that the current time is outside the officer's/resource's office hours

When the page loads

Then I see "Office closed" with the next open time

And I see an emergency or hotline fallback number

US-08 [Added after GenAI]:

“As a veteran, I want event flyers to be downloadable so that I can print or share them offline.”

Acceptance Scenarios (Gherkin):

Positive Scenario: Download event flyer as PDF

Given that I am viewing an event details page

And the event flyer is visible on the page

When I click the “Download Flyer (PDF)” button

Then the system generates a PDF of the flyer

And the file begins downloading to my device

And the downloaded file opens successfully and is readable

Negative/Edgecase Scenario: Download fails

Given that I am viewing an event details page

When I click the “Download Flyer (PDF)” button

And the file cannot be generated due to a server error

Then I see an error message reading “Download currently unavailable. Please try again later.”

And I see a link or button labeled “Report an issue”

US-09 [Added after GenAI]:

“As an admin, I want past events to be archived automatically so that I do not have to manually maintain the event list.”

Acceptance Scenarios (Gherkin):

Positive Scenario: Automatic archival after event end date

Given that I am authenticated as an admin

And an event ended yesterday at 11:59 PM local time

When the page loads

Then the event no longer appears in the “Upcoming Events” section

And the event appears in the “Past Events” section with its original flyer and details

And the event remains searchable via the site search

Negative/Edgecase Scenario: Event spans multiple days and is still ongoing

Given that I am authenticated as an admin

And an event started yesterday and ends tomorrow at 5:00 PM local time

When the page loads today

Then the event still appears in the “Upcoming Events” section

And it does not appear in “Past Events”

And I see the event labeled with the correct ongoing date range (e.g., “Sept 20–22”)

4. Brainstorming and GenAI Reflection

Step A: Team Brainstorming

After a thorough discussion, we’ve decided there are 6 key decisions that we’d like to emphasize:

- Decision Point 1 (FR): **FR-02** is a key functional decision.
- Decision Point 2 (FR): **FR-03** is a key functional decision.
- Decision Point 3 (NFR): **NFR-01** is a key non-functional decision.
- Decision Point 4 (NFR): **NFR-03** is a key non-functional decision.
- Decision Point 5 (US): **US-01** is a key user story decision.
- Decision Point 6 (US): **US-05** is a key user story decision.

Step B: GenAI-Assisted Brainstorming

(Summarize what GenAI suggested — new FRs, NFRs, user stories, or scenarios.)

Through an iterative process of prompting the AI, the team was able to get some more functional and non-functional requirements, as well as 2 new user stories. The GenAI used was ChatGPT 5, and both this document and our team's work up until this point, along with our project description document, were uploaded to be reviewed by the AI. Below is

the prompting that was used in order to get a more in-depth review of our current work and refine it for further use.

Prompt 1: *"Look over both documents. The Project Description gives an overview of our project. Assignment 1 covers our NFR + FR requirements, user stories, and scenarios with Gherkin code. Please analyze both documents and complete step 4 parts b and c of WCVS_Assignment1_Requirments. Try to look for anything we should add or refine. You can provide feedback and insights or new material for functional requirements, non-functional requirements, user stories, or scenarios."*

Prompt 2: *"Can you review our functional and non-functional requirements to see if there are any common gaps in resource/event platforms that we might have overlooked? We want practical but realistic additions that align with the project scope."*

Prompt 3: *"Looking at our user stories and acceptance scenarios, can you suggest ways to improve accessibility and reliability? We want to make sure older veterans or less tech-savvy users still have a good experience."*

Prompt 4: *"Now that we've covered basic features, can you propose refinements that support admins and long-term sustainability of the website? Think about accountability, automation, and content management."*

- GenAI Suggestion 1:
FR-09 [New]: The system shall include a search function that allows users to quickly locate resources, events, or contact information by keyword.
- GenAI Suggestion 2:
FR-10 [New]: The system shall allow event flyers to be downloadable as PDFs for offline use.
- GenAI Suggestion 3:
FR-11 [New]: The system shall automatically archive past events while keeping them accessible under a "Past Events" section.
- GenAI Suggestion 4:
NFR-09 [New]: The system shall achieve a page load time of ≤ 3 seconds on standard broadband connections to ensure accessibility for all users.

- GenAI Suggestion 5:
NFR-10 [New]: The system shall maintain 99% uptime monthly to ensure reliability for veterans seeking time-sensitive resources.
- GenAI Suggestion 6:
NFR-11 [New]: The system shall ensure that all administrative actions (e.g., event creation) are logged for accountability and auditing.
- GenAI Suggestion 7:
US-07 [New]:

“As a veteran, I want event flyers to be downloadable so that I can print or share them offline.”

Acceptance Scenarios (Gherkin):

Positive Scenario: Download event flyer as PDF

Given that I am viewing an event details page
 And the event flyer is visible on the page
 When I click the “Download Flyer (PDF)” button
 Then the system generates a PDF of the flyer
 And the file begins downloading to my device
 And the downloaded file opens successfully and is readable

Negative/Edgecase Scenario: Download fails

Given that I am viewing an event details page
 When I click the “Download Flyer (PDF)” button
 And the file cannot be generated due to a server error
 Then I see an error message reading “Download currently unavailable. Please try again later.”
 And I see a link or button labeled “Report an issue”

- GenAI Suggestion 8:
US-08 [New]:

“As an admin, I want past events to be archived automatically so that I do not have to manually maintain the event list.”

Acceptance Scenarios (Gherkin):

Positive Scenario: Automatic archival after event end date

Given that I am authenticated as an admin
 And an event ended yesterday at 11:59 PM local time
 When the page loads
 Then the event no longer appears in the “Upcoming Events” section

And the event appears in the “Past Events” section with its original flyer and details
And the event remains searchable via the site search

Negative/Edgecase Scenario: Event spans multiple days and is still ongoing

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Step C: Refined Requirements & Stories

(Update Step A with insights from GenAI. Clearly mark new/modified items — e.g., FR-06 [Added after GenAI], US-04 [Modified after GenAI])

This was completed; all updates can be seen in the sections from step A with the correct tag attached to them.

Step D: Reflection (200–300 words)

- How did you feel about using GenAI in this exercise (e.g., empowering, surprising, confusing, over-reliant)?

We found using GenAI to be extremely helpful and empowering. It allowed us to brainstorm new ideas more effectively, refine and improve our plans and agendas, and paraphrase our writing so that our documentation became more structured, clear, and cohesive.

- In what ways did GenAI change or improve your brainstorming compared to your team’s initial work? (Consider clarity, creativity, and coverage.)

GenAI significantly improved our brainstorming process by making our ideas more organized and comprehensive. It helped us clarify vague thoughts and turn them into manageable and actionable plans. It also encouraged creativity by suggesting new perspectives we had not considered, and ensured broader coverage by filling in gaps we may have missed. Overall, it is a great tool for us to stay productive, focused, and innovative.

- Did GenAI help you uncover new functional, non-functional, or business requirements that you had not considered before? Provide examples.

Yes, the use of GenAI did help us to uncover new requirements that we had not considered before. An example of this is **FR-09**, the team had not considered the ability to download the event flyers for offline use. Another example can be seen in **NFR-11**, which covers logging all system changes, such as events created by the admin, to track updates to the page. GenAI made it much more effective to think of small details that are easy to overlook.